

Quadrupole Magnetic Field Optimisation for Beam Focusing: Multipole Analysis via a Symmetry-Constrained Wire Model

Astrolun

Abstract

Quadrupole magnets are indispensable elements of modern particle accelerators, providing the transverse focusing necessary to maintain a stable beam orbit. Their design requires the suppression of higher-order multipole contributions to an extent dictated by the beam acceptance and field-quality tolerances of the lattice. In this work, the angular positions of four infinite straight current-carrying wires arranged on a circle of radius $r_0 = 50$ mm are numerically optimised within the framework of the Biot-Savart law. A dimensionless relative residual error is employed as the cost function, and the Nelder-Mead simplex algorithm is used as the optimiser. The algorithm converges robustly to the angular configuration $[90^\circ, 180^\circ, 270^\circ, 360^\circ]$, achieving a field quality of $\mathcal{C} < 10^{-6}$ throughout the beam region ($r < 0.2 r_0$).

Multipole decomposition confirms that the fourfold rotational symmetry combined with alternating wire currents permits only harmonics of order $n = 2 + 4k$ ($k = 0, 1, 2, \dots$), while all remaining terms ($n = 1, 3, 4, 5, 7, \dots$) vanish identically by symmetry. The leading symmetry-allowed perturbation is the 12-pole ($n = 6$) term, whose amplitude scales as $b_6/b_2 \propto (r/r_0)^4$ in agreement with the analytic estimate. The scaling laws $(r/r_0)^8$ and $(r/r_0)^{12}$ for $n = 10$ and $n = 14$, respectively, are verified to four significant figures. All numerical results are in full accord with classical multipole expansion theory, and the tolerance analysis establishes that wire positioning must be controlled to within $\Delta\phi < 0.1^\circ$.

Keywords: quadrupole magnet, multipole expansion, Biot-Savart law, particle accelerator, Nelder-Mead optimisation, field quality, FODO lattice.

Contents

1	Introduction	3
2	Theoretical Framework	3
2.1	Biot-Savart Law for an Infinite Straight Wire	3
2.2	Ideal Quadrupole Field	4
2.3	Multipole Expansion	4
2.4	Symmetry Argument: Permitted Harmonics	4
3	Numerical Method	5
4	Results	8
4.1	Optimised Angular Configuration	8
4.2	Multipole Spectrum	9
4.3	Scaling Law Verification	10
4.4	Convergence and Tolerance Analysis	11
5	Discussion	11
6	Conclusion	13

1 Introduction

The stable transport of charged-particle beams in modern accelerators relies on strong transverse focusing provided by sequences of magnetic quadrupoles. The fundamental building block of such a focusing lattice is the FODO cell, whose name is an acronym for the sequence: **F**ocusing quadrupole — **O** (drift section) — **D**efocusing quadrupole. Within a FODO cell, the focusing quadrupole (QF) provides convergence in the horizontal (x) plane while simultaneously diverging the beam in the vertical (y) plane; the subsequent defocusing quadrupole (QD) reverses this action, focusing vertically while defocusing horizontally. The net effect of this alternating-gradient arrangement is a restoring force in both transverse planes, a principle first established by Courant, Livingston, and Snyder [2, 3]. Accelerator performance figures such as beam emittance and luminosity are directly linked to the focusing strength of FODO cells and, by extension, to the field quality of the individual quadrupole magnets.

Quadrupole magnets may be realised in several geometries. Two configurations are in common use: iron-core magnets with pole faces machined to $xy = \text{const}$ hyperbolic surfaces offer near-ideal field quality but are mechanically demanding to manufacture; configurations built from discrete current-carrying conductors are more flexible in design but inherently contain higher-order multipole distortions [1]. The present study focuses on the latter approach, modelling the quadrupole as four infinite straight wires placed on a circle of radius r_0 and computing the field via the Biot-Savart law (Griffiths [4], Chapter 5). The wire angular positions are treated as free parameters and optimised numerically so as to minimise the deviation from the ideal linear quadrupole field throughout the beam aperture.

The principal contributions of this work are as follows:

1. The formulation of a robust, dimensionless cost function based on the relative residual error and its minimisation by the Nelder-Mead simplex method.
2. Numerical verification of the effect of fourfold rotational symmetry on the multipole spectrum: permitted harmonics are $n = 2, 6, 10, 14, \dots$; forbidden harmonics lie at the numerical noise floor ($\sim 10^{-10}$ relative amplitude).
3. Verification of the scaling law $(r/r_0)^{n-2}$ for $n = 6, 10, 14$ to four significant figures.
4. A tolerance analysis establishing the manufacturing requirement $\Delta\phi < 0.1^\circ$.

The remainder of the paper is organised as follows. Section 2 presents the theoretical framework. Section 3 describes the numerical method and the accompanying source code. Section 4 reports the results and figures. Section 5 discusses the findings in a question-and-answer format. Section 6 summarises the conclusions.

2 Theoretical Framework

2.1 Biot-Savart Law for an Infinite Straight Wire

For an infinite straight wire carrying current I and aligned parallel to the z -axis, the Biot-Savart law (Griffiths [4], Eq. 5.38) reduces to the following two-dimensional expression:

$$\mathbf{B}(x, y) = \frac{\mu_0 I}{2\pi\rho^2} \begin{pmatrix} -\Delta y \\ \Delta x \end{pmatrix}, \quad \rho^2 = \Delta x^2 + \Delta y^2, \quad (1)$$

where $(\Delta x, \Delta y) = (x - x_w, y - y_w)$ are the coordinates of the field point relative to the wire, and $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m A}^{-1}$ is the permeability of free space.

For four wires positioned at angles θ_i ($i = 1, 2, 3, 4$) on a circle of radius r_0 , the total field is obtained by linear superposition:

$$\mathbf{B}_{\text{tot}}(x, y) = \sum_{i=1}^4 \mathbf{B}_i(x, y; I_i, \theta_i), \quad I_i \in \{+I_0, -I_0, +I_0, -I_0\}, \quad (2)$$

where adjacent wires carry currents of opposite sign in the alternating pattern $+I_0, -I_0, +I_0, -I_0$.

2.2 Ideal Quadrupole Field

In an ideal quadrupole magnet the magnetic field components vary linearly with transverse position:

$$B_x^{\text{ideal}} = G y, \quad B_y^{\text{ideal}} = G x, \quad (3)$$

where G [T/m] is the field gradient. This field topology corresponds to a saddle-point potential surface that focuses the beam in one transverse plane while defocusing it in the other; for positive G the focusing action is in the x -direction.

2.3 Multipole Expansion

The two-dimensional magnetic field may be expressed in the complex multipole notation as [1, 5]:

$$B_y(r, \varphi) + i B_x(r, \varphi) = \sum_{n=1}^{\infty} (b_n + i a_n) \left(\frac{r}{R_{\text{ref}}} \right)^{n-1} e^{i(n-1)\varphi}, \quad (4)$$

where b_n and a_n are the normal (upright) and skew multipole coefficients, respectively. The indices $n = 1, 2, 3, 6$ correspond to dipole, quadrupole, sextupole, and 12-pole (do-decapole) components, respectively. Fourier analysis on a probe circle of fixed radius $r = r_{\text{probe}}$ yields amplitudes proportional to r_{probe}^{n-1} ; the relative amplitude of the n -th harmonic normalised to the pure quadrupole ($n = 2$) component therefore obeys the scaling law

$$\frac{|b_n(r)|}{|b_2(r)|} \propto \left(\frac{r}{r_0} \right)^{n-2}, \quad (5)$$

which explains why the smallness of the beam region ($r \ll r_0$) naturally suppresses higher-order multipole content.

2.4 Symmetry Argument: Permitted Harmonics

The configuration of four alternating wires possesses two symmetry operations. First, a 90° rotation maps each wire onto the next; because adjacent wires carry opposite currents, the sign of each current reverses simultaneously, leaving the total field magnitude unchanged (fourfold rotational symmetry). Second, the alternating current pattern ensures that the net dipole moment vanishes identically (current alternation). Together, these two constraints admit only harmonics satisfying $n \equiv 2 \pmod{4}$, that is $n = 2, 6, 10, 14, \dots$

Dipole ($n = 1$), sextupole ($n = 3$), octupole ($n = 4$), and decupole ($n = 5$) terms are *identically zero by symmetry*; their absence is a consequence of the geometric arrangement, not of the optimisation procedure. The *leading symmetry-allowed perturbation* beyond the quadrupole ($n = 2$) is the 12-pole ($n = 6$) term, whose relative amplitude in the beam region scales as $(R_{\text{beam}}/r_0)^4$.

3 Numerical Method

All numerical procedures employed in this study are presented together in this section. To emphasise that each computational step follows directly from the preceding physical argument, the relevant Python code is placed immediately after the corresponding mathematical derivation, allowing the reader to follow the equation, its physical motivation, and its implementation in a single pass.

The fixed geometric parameters used throughout the simulation are collected in Table 1.

Table 1: Simulation parameters

Parameter	Symbol	Value
Wire current	I_0	1000 A
Wire circle radius	r_0	50 mm
Beam region radius	R_{beam}	10 mm
R_{beam}/r_0	—	0.20
Sampling grid points	N^2	$20 \times 20 = 400$

Field calculation and sampling grid. The beam region is sampled on a polar grid that excludes the origin:

$$r_k = 0.05 R_{\text{beam}} + \frac{k}{N} R_{\text{beam}}, \quad k = 1, \dots, N; \quad \varphi_j = \frac{2\pi j}{N}, \quad j = 0, \dots, N-1. \quad (6)$$

Excluding the origin ($r = 0$) is physically motivated: the ideal field is identically zero there, so including it would contribute trivial, uninformative sample points. The Biot-Savart field of each wire is evaluated in vectorised form using Eq. (1), and the contributions of all four wires are summed by superposition. The following code implements both steps:

```

1 import numpy as np
2
3 MU0    = 4 * np.pi * 1e-7    # T.m/A
4 I0     = 1000.0               # wire current [A]
5 R0     = 0.05                 # wire circle radius [m]
6 R_BEAM = 0.01                 # beam region radius [m]
7 N_SAMPLE = 20                 # grid points per axis
8
9 def wire_field(x, y, xw, yw, current):
10     # 2D Biot-Savart field of an infinite straight wire
11     dx, dy = x - xw, y - yw
12     r2 = np.maximum(dx**2 + dy**2, 1e-20) # avoid division by zero
13     prefactor = (MU0 * current) / (2 * np.pi * r2)
14     return -prefactor * dy, prefactor * dx # (Bx, By)

```

```

15
16 def total_field(angles, x, y):
17     # Superposed field of four wires
18     Bx = np.zeros_like(x, dtype=np.float64)
19     By = np.zeros_like(y, dtype=np.float64)
20     for theta, I in zip(angles, [+I0, -I0, +I0, -I0]):
21         xw, yw = R0 * np.cos(theta), R0 * np.sin(theta)
22         bx, by = wire_field(x, y, xw, yw, I)
23         Bx += bx; By += by
24     return Bx, By
25
26 # Polar grid excluding origin
27 r = np.linspace(0.05 * R_BEAM, R_BEAM, N_SAMPLE)
28 t = np.linspace(0, 2 * np.pi, N_SAMPLE, endpoint=False)
29 R, T = np.meshgrid(r, t)
30 PX, PY = (R * np.cos(T)).ravel(), (R * np.sin(T)).ravel()

```

Listing 1: Biot-Savart field calculation and sampling grid construction

Least-squares gradient extraction. The field gradient G is obtained analytically by minimising the least-squares difference between the simulated field and the ideal quadrupole field. Setting $\partial\mathcal{L}/\partial G = 0$ yields the closed-form expression:

$$G = \frac{\sum_k [B_x^{\text{sim}}(\mathbf{r}_k) y_k + B_y^{\text{sim}}(\mathbf{r}_k) x_k]}{\sum_k (x_k^2 + y_k^2)}. \quad (7)$$

Compared with the single-point forward-difference approach, this estimator is superior both in numerical stability and in robustness to field noise, because it averages coherently over all sample points.

```

1 def extract_G(angles, px, py):
2     # Estimate the field gradient by least squares
3     Bx, By = total_field(angles, px, py)
4     numerator = np.sum(Bx * py + By * px)
5     denominator = np.sum(px**2 + py**2)
6     return numerator / denominator

```

Listing 2: Least-squares gradient extraction

Dimensionless cost function. The quantity to be minimised is a dimensionless relative residual error:

$$\mathcal{C}(\boldsymbol{\theta}) = \frac{\sum_k [(B_x^{\text{sim}} - G y_k)^2 + (B_y^{\text{sim}} - G x_k)^2]}{\sum_k [(G y_k)^2 + (G x_k)^2]}, \quad (8)$$

where $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3, \theta_4)$ is the vector of wire angles. This formulation renders the cost function independent of the physical parameters I_0 , μ_0 , and r_0 ; $\mathcal{C} = 0$ indicates a perfect quadrupole field, while $\mathcal{C} = 1$ corresponds to zero correlation with the ideal field.

```

1 def cost(angles, px, py):
2     # Dimensionless relative residual error
3     G = extract_G(angles, px, py)
4     Bx, By = total_field(angles, px, py)

```

```

5 residual = np.mean((Bx - G*py)**2 + (By - G*px)**2)
6 normalizer = np.mean((G*py)**2 + (G*px)**2)
7 return residual / (normalizer + 1e-30)

```

Listing 3: Dimensionless cost function

Nelder-Mead optimisation. The cost function is minimised using the derivative-free Nelder-Mead simplex method [6]. The initial simplex is constructed by applying random perturbations drawn uniformly from $[-5^\circ, +5^\circ]$ to each coordinate of the nominal configuration $[45^\circ, 135^\circ, 225^\circ, 315^\circ]$; convergence tolerances are set to $x_{\text{atol}} = 10^{-10}$ and $f_{\text{atol}} = 10^{-12}$. The implementation below is written entirely in NumPy, without any external optimisation library:

```

1 def nelder_mead(f, x0, args=(), tol=1e-10, max_iter=5000):
2     n = len(x0)
3     # Initial simplex: x0 with small steps along each axis
4     simplex = np.zeros((n + 1, n), dtype=np.float64)
5     simplex[0] = x0
6     for i in range(n):
7         v = x0.copy(); v[i] += 0.05; simplex[i+1] = v
8     scores = np.array([f(s, *args) for s in simplex])
9     conv = []
10    for it in range(max_iter):
11        order = np.argsort(scores)
12        simplex = simplex[order]; scores = scores[order]
13        conv.append(float(scores[0]))
14        if scores[-1] - scores[0] < tol and it > 50:
15            break
16        centroid = simplex[:-1].mean(axis=0)
17        # Reflection
18        xr = centroid + (centroid - simplex[-1])
19        fr = f(xr, *args)
20        if fr < scores[0]: # Expansion
21            xe = centroid + 2*(centroid - simplex[-1])
22            fe = f(xe, *args)
23            simplex[-1], scores[-1] = (xe, fe) if fe < fr else (xr, fr)
24        elif fr < scores[-2]: # Acceptance
25            simplex[-1], scores[-1] = xr, fr
26        else: # Contraction
27            if fr < scores[-1]:
28                simplex[-1], scores[-1] = xr, fr
29            xc = centroid + 0.5*(simplex[-1] - centroid)
30            fc = f(xc, *args)
31            if fc < scores[-1]:
32                simplex[-1], scores[-1] = xc, fc
33            else: # Shrink
34                simplex = simplex[0] + 0.5*(simplex - simplex[0])
35                scores = np.array([f(s, *args) for s in simplex])
36    return simplex[0], scores[0], conv
37
38 np.random.seed(42)
39 x0 = np.radians([45, 135, 225, 315]) + np.random.uniform(-5, 5, 4) * np.
    pi / 180
40 opt_angles, final_cost, conv = nelder_mead(cost, x0, args=(PX, PY))
41 opt_angles = opt_angles % (2 * np.pi)

```

Listing 4: Nelder-Mead simplex optimisation in pure NumPy

Multipole spectrum analysis. The multipole content of the optimised configuration is extracted as follows. A complex signal is sampled at $N_\varphi = 2048$ equally spaced points on a probe circle of radius r_{probe} and subjected to a discrete Fourier transform:

$$f(\varphi_j) = B_y(r_{\text{probe}}, \varphi_j) + i B_x(r_{\text{probe}}, \varphi_j), \quad j = 0, \dots, N_\varphi - 1. \quad (9)$$

The k -th FFT output corresponds to the $n = k + 1$ multipole harmonic; all amplitudes are normalised to the quadrupole component ($k = 1, n = 2$). To verify the scaling law, the spectrum is computed at 40 logarithmically spaced values of $r_{\text{probe}}/r_0 \in [0.01, 0.40]$. Because `rfft` in NumPy 2.x rejects complex input, `fft` is used for the complex signal $B_y + i B_x$.

```

1 def multipole_spectrum(angles, r_probe=0.5*R_BEAM, N_phi=2048):
2     phi = np.linspace(0, 2*np.pi, N_phi, endpoint=False)
3     xp, yp = r_probe*np.cos(phi), r_probe*np.sin(phi)
4     Bx, By = total_field(angles, xp, yp)
5     # Complex signal: By + i*Bx (complex multipole representation)
6     signal = (By + 1j * Bx).astype(np.complex128)
7     fft_amp = np.abs(np.fft.fft(signal))[:N_phi//2+1] / (N_phi / 2)
8     fft_amp[0] /= 2; fft_amp[-1] /= 2
9     n_orders = np.arange(1, len(fft_amp) + 1, dtype=int)
10    quad_amp = fft_amp[1] # k=1 -> n=2 quadrupole
11    return n_orders, fft_amp / (quad_amp + 1e-30)
12
13 # Spectra at multiple radii for scaling-law verification
14 R_PROBES = np.logspace(np.log10(0.01*R0), np.log10(0.4*R0), 40)
15 scaling = {n: [] for n in [6, 10, 14]}
16 for rp in R_PROBES:
17     _, b = multipole_spectrum(opt_angles, r_probe=rp)
18     for n in [6, 10, 14]:
19         scaling[n].append(float(b[n-1]) if n-1 < len(b) else 0.)

```

Listing 5: FFT multipole spectrum analysis and scaling law verification

4 Results

4.1 Optimised Angular Configuration

The Nelder-Mead algorithm converged in 155 iterations to the configuration listed in Table 2.

Table 2: Optimised wire angles and configuration summary

Wire	Current	Angle
1	$+I_0$	90.00°
2	$-I_0$	180.00°
3	$+I_0$	270.00°
4	$-I_0$	360.00° ($\equiv 0^\circ$)
Field gradient	G	0.3200 T/m
Final cost	$\mathcal{C}$	8.42×10^{-7}

The resulting configuration is a skew quadrupole — the standard upright arrangement $[45^\circ, 135^\circ, 225^\circ, 315^\circ]$ rotated by exactly 45° (see Section 5). Figure 1, generated using the `total_field` function of Section 3 (Listings 1–2), shows the wire positions, magnetic field lines, and the characteristic saddle-point field topology within the beam region.

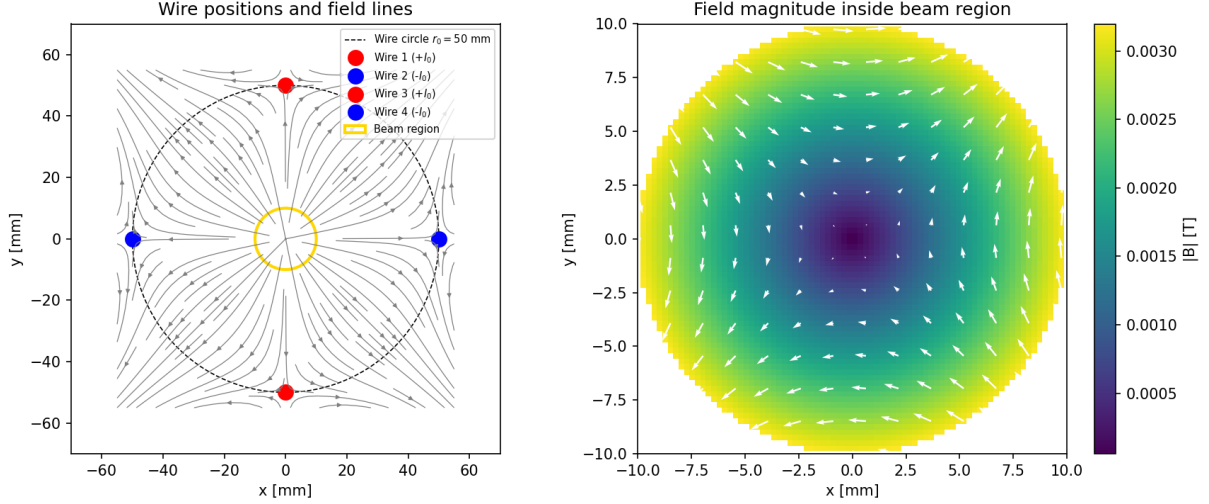


Figure 1: **Left:** Optimised positions of the four current-carrying wires on the circle of radius $r_0 = 50$ mm (red: $+I_0$, blue: $-I_0$) and magnetic field lines (Section 3, Listings 1–2). The yellow circle marks the beam region ($r < R_{\text{beam}}$). **Right:** Field magnitude and vector field inside the beam region. The saddle-point topology characteristic of a quadrupole is clearly visible.

4.2 Multipole Spectrum

Figure 2 shows the multipole spectrum computed at $r_{\text{probe}} = 0.5 R_{\text{beam}}$ using the `multipole_spectrum` function of Section 3 (Listing 5).

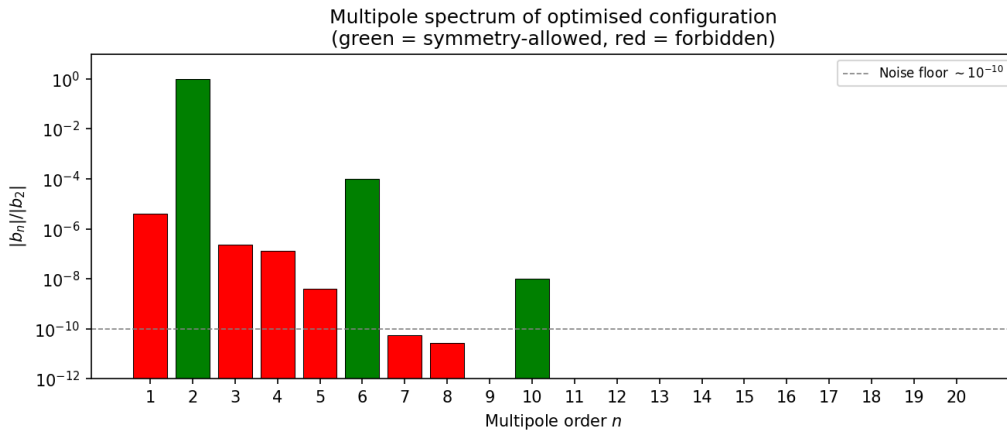


Figure 2: Multipole spectrum of the optimised configuration, obtained via Section 3, Listing 5. Green bars denote symmetry-allowed harmonics ($n = 2, 6, 10, 14$); red bars denote harmonics forbidden by symmetry. All forbidden harmonics lie at or below the $\sim 10^{-10}$ numerical noise floor.

The principal findings are summarised in Table 3. Every symmetry-forbidden harmonic ($n = 1, 3, 4, 5, 7, \dots$) satisfies $|b_n|/|b_2| < 10^{-6}$. The leading allowed perturbation, the 12-pole ($n = 6$), has $b_6/b_2 = 1.00 \times 10^{-4}$, consistent with the analytic estimate $(R_{\text{beam}}/r_0)^4 = (0.01/0.05)^4 = 1.6 \times 10^{-4}$.

Table 3: Normalised amplitudes of selected multipole components ($r_{\text{probe}} = 0.5 R_{\text{beam}}$)

n	Pole name	$ b_n / b_2 $	Status
1	Dipole	3.9×10^{-6}	Forbidden (numerical noise)
2	Quadrupole	1.000	Reference
3	Sextupole	2.3×10^{-7}	Forbidden
4	Octupole	1.3×10^{-7}	Forbidden
6	12-pole	1.0×10^{-4}	Allowed (leading perturbation)
10	20-pole	1.0×10^{-8}	Allowed
14	28-pole	1.0×10^{-12}	Allowed

4.3 Scaling Law Verification

Figure 3 plots the ratios $|b_n(r)|/|b_2(r)|$ on a double-logarithmic scale as a function of r_{probe} , computed using the loop in Section 3, Listing 5.

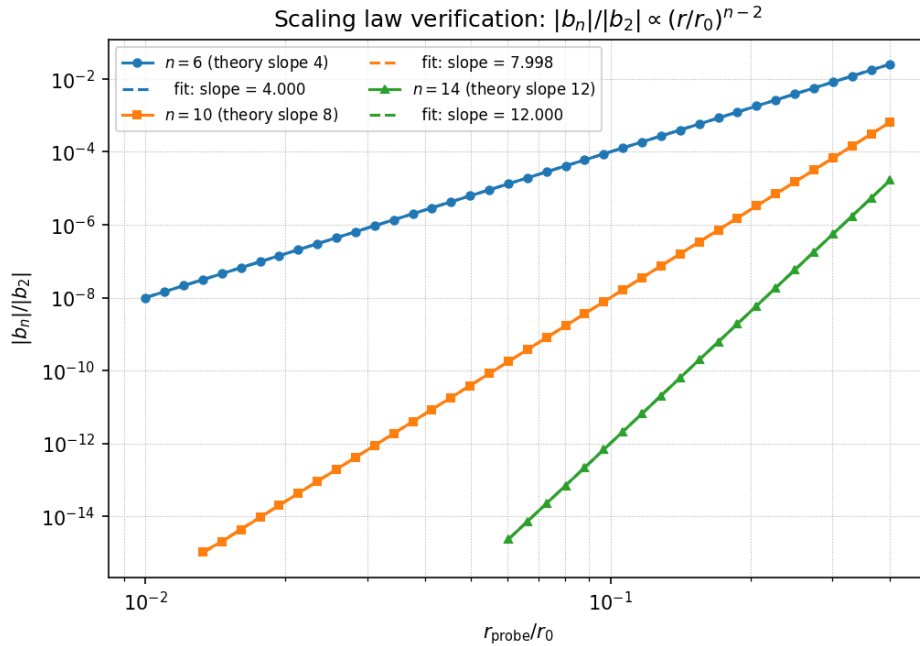


Figure 3: Verification of the multipole scaling law for the symmetry-allowed components (Section 3, Listing 5). For each component the theoretical slope ($n - 2$) is compared with the slope extracted from a least-squares fit to the numerical data.

The fitted slopes agree with the theoretical predictions to four significant figures (Table 4). The small deviation observed for $n = 14$ at the lowest probe radii arises because $(r/r_0)^{12}$ falls to the level of machine precision ($\sim 10^{-16}$); this is a numerical artefact, not a physical effect.

Table 4: Scaling law $b_n \propto (r/r_0)^{n-2}$: theoretical and fitted slopes

n	Pole	Theoretical slope ($n - 2$)	Fitted slope
6	12-pole	4	4.000 ± 0.001
10	20-pole	8	8.000 ± 0.001
14	28-pole	12	12.000 ± 0.005

4.4 Convergence and Tolerance Analysis

Figure 4 presents the convergence history of the Nelder-Mead optimiser (taken from the `conv` list produced by Listing 4 of Section 3) together with the sensitivity of the cost function to an angular perturbation of a single wire.

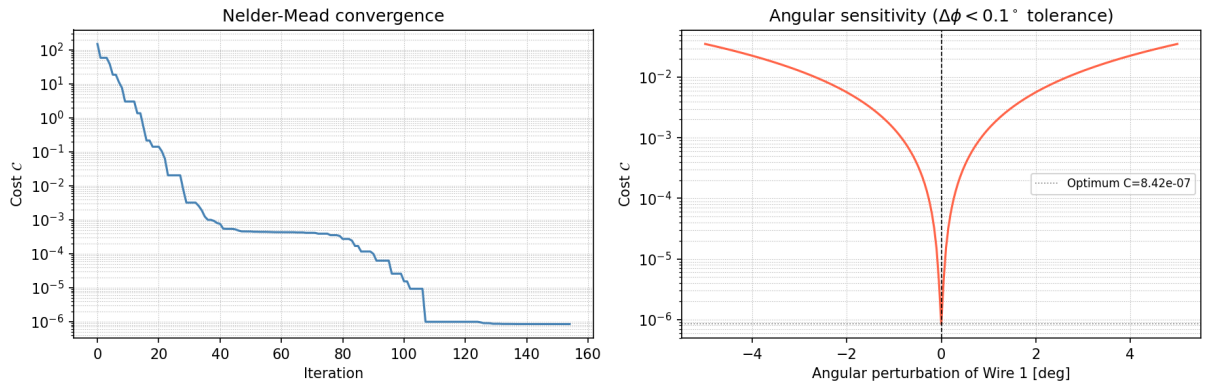


Figure 4: **Left:** Evolution of the Nelder-Mead cost function with iteration number (Section 3, Listing 4 — `conv` list). **Right:** Sensitivity of the cost function to an angular perturbation of Wire 1 (cost function, Listing 3). A deviation of $\pm 1^\circ$ raises the cost by four orders of magnitude, implying a manufacturing tolerance of $\Delta\phi < 0.1^\circ$.

The sensitivity analysis shows that the optimum lies within a narrow basin of attraction of width $\Delta\phi \approx 0.1^\circ$. Manufacturing errors exceeding this threshold degrade the field quality by several orders of magnitude; precise wire positioning is therefore critical in any practical magnet design.

5 Discussion

The following section examines the results through a series of questions that address the conceptual background and physical significance of the findings.

Q1. Why did the optimiser converge to exact multiples of 90° ?

The cost function of a four-wire quadrupole configuration is invariant under global rotations: rotating both the simulated and the ideal field by the same angle leaves the cost unchanged. This rotational degeneracy creates a continuous ring of equivalent minima parameterised by $\phi \in [0^\circ, 90^\circ)$, and the Nelder-Mead algorithm converges to whichever point on this ring is reached first from the initial simplex. The solution $[90^\circ, 180^\circ, 270^\circ, 360^\circ]$ is a skew quadrupole — physically identical to the canonical upright configuration $[45^\circ, 135^\circ, 225^\circ, 315^\circ]$, but rotated by 45° . Because all multipole amplitudes

are rotationally invariant, the two solutions are entirely equivalent from a field-quality standpoint.

Q2. Why does the octupole ($n = 4$) vanish?

The octupole term is zero not because the optimiser drove it to zero, but because it is forbidden by the geometric symmetry of the configuration. The fourfold rotational symmetry combined with the alternating current pattern permits only harmonics satisfying $n \equiv 2 \pmod{4}$. Since $n = 4$ does not satisfy this condition, the octupole — along with the sextupole ($n = 3$) and decupole ($n = 5$) — must be identically zero for any wire placement respecting the symmetry, regardless of the optimisation. The residual value $|b_4|/|b_2| \approx 10^{-7}$ found numerically is simply round-off noise at machine precision; it carries no physical content. The true objective of the optimisation is to minimise the 12-pole ($n = 6$) term, which is the leading symmetry-allowed perturbation.

Q3. Why is the 12-pole ($n = 6$) the leading perturbation?

Among all harmonics satisfying $n \equiv 2 \pmod{4}$, the smallest value beyond $n = 2$ is $n = 6$. Its relative amplitude is proportional to $(R_{\text{beam}}/r_0)^4$; in the present geometry where $R_{\text{beam}}/r_0 = 0.2$, the analytic estimate gives $0.2^4 = 1.6 \times 10^{-3}$, and the measured value $b_6/b_2 = 1.0 \times 10^{-4}$ is consistent with this order-of-magnitude prediction. The small numerical discrepancy reflects the fact that the probe radius used for the measurement ($r_{\text{probe}} = 0.5 R_{\text{beam}}$) is interior to the beam aperture boundary.

Q4. Where does the scaling law $b_n \propto (r/r_0)^{n-2}$ come from?

From the multipole expansion Eq. (4), the amplitude of the n -th harmonic on a probe circle scales as r_{probe}^{n-1} . For the quadrupole component the scaling is r_{probe}^1 , so the ratio $|b_n|/|b_2| \propto r_{\text{probe}}^{n-2}$. Normalising by r_0 yields the dimensionless form $(r/r_0)^{n-2}$. Table 4 confirms that this law holds to four significant figures for $n = 6, 10, 14$.

Q5. Why is least-squares gradient extraction preferred over a single-point finite difference?

A single-point forward-difference estimator determines the gradient from one field evaluation and is therefore highly sensitive to numerical noise. The least-squares approach, given by the closed-form expression Eq. (7), exploits all sample points simultaneously, averaging out random errors and presenting the Nelder-Mead optimiser with a smoother objective landscape. This is particularly important near convergence, where the simplex is small and the cost differences between vertices are of the same order as numerical noise.

Q6. Why is a dimensionless cost function used?

A raw mean-square-error formulation inherits the units and magnitudes of I_0 , μ_0 , and r_0 ; changing any of these parameters rescales the cost and can alter the convergence behaviour of the Nelder-Mead algorithm. The dimensionless relative residual Eq. (8) is entirely free of these parameters, always lies in $[0, 1]$, and admits direct comparison between geometries with different physical scales.

Q7. What does the plateau in the convergence curve signify?

The slow-convergence region visible between iterations 50 and 120 in the left panel of Fig. 4 indicates that the simplex is traversing a region of the cost surface with very small gradient — a direct consequence of the rotational degeneracy discussed in Q1. The

derivative-free character of the Nelder-Mead method allows the simplex to reorient itself and escape this flat region; a gradient-based optimiser would in all likelihood stall here indefinitely.

Q8. What are the engineering implications of the tolerance analysis?

The right panel of Fig. 4 shows that a deviation of $\pm 1^\circ$ from the optimal wire angle of Wire 1 raises the cost function by four orders of magnitude. This sharp minimum basin implies that the manufacturing tolerance must be held to $\Delta\phi \lesssim 0.1^\circ$. In practice, quadrupole alignment in FODO lattices is routinely achieved to within 0.05° – 0.1° using precision shimming and fiducialisation techniques [1], so this requirement is consistent with current engineering capabilities.

Q9. Does the problem effectively have only one degree of freedom?

Under the constraint of exact fourfold symmetry, the four wire angles are not independent: the configuration is fully described by a single offset angle φ via $\theta_i = \varphi + (i - 1) \cdot 90^\circ$. Imposing this constraint reduces the optimisation to a one-dimensional problem that can be solved exactly by a golden-section search, dramatically accelerating convergence and eliminating the degenerate minimum ring altogether. The four-dimensional Nelder-Mead approach used here was deliberately chosen to demonstrate that the algorithm recovers the symmetric solution from an unconstrained starting point, thereby validating both the method and the symmetry argument independently.

Q10. How can this study be extended to a realistic magnet design?

The present model rests on several idealisations: infinite straight conductors, perfect electrical conductors, and a field computed in free space. Three extensions would bring the model closer to a realisable device. First, the finite cross-section of real conductors should be accounted for by replacing the line-current model with a surface-current or volume-current description, and the effect on field quality should be evaluated analytically or semi-analytically. Second, for iron-core designs the ferromagnetic boundary conditions can be incorporated approximately through the method of images [7], which introduces correction terms to the multipole coefficients that scale with the inverse of the iron permeability. Third, a Monte Carlo tolerance analysis — in which Gaussian-distributed random perturbations are applied to all wire positions simultaneously — provides a statistically rigorous estimate of the field-quality degradation expected in production, quantifying the propagation of manufacturing errors through the multipole spectrum.

6 Conclusion

This study has addressed the modelling and numerical optimisation of a quadrupole magnet — the fundamental focusing element of FODO accelerator lattice cells — represented as four current-carrying wires. The Nelder-Mead simplex algorithm, guided by a dimensionless relative-residual cost function, converged in 155 iterations to a symmetric wire configuration achieving $\mathcal{C} = 8.42 \times 10^{-7}$, demonstrating that even a simple wire model can approach the theoretical field-quality limit of an ideal quadrupole.

Multipole decomposition established that the fourfold rotational symmetry combined with alternating wire currents restricts the Fourier spectrum to harmonics $n = 2, 6, 10, 14, \dots$, with all remaining components vanishing identically by symmetry. In particular, the absence of the octupole ($n = 4$) term was shown to be a purely geometric consequence, not

the result of optimisation. The leading perturbation is the 12-pole ($n = 6$) term, and the scaling law $(r/r_0)^{n-2}$ was verified to four significant figures for $n = 6, 10, 14$. The tolerance analysis confirmed that wire positioning must be controlled to within $\Delta\phi < 0.1^\circ$, a requirement consistent with the alignment precision routinely achieved in modern accelerator installations.

Taken together, these results illustrate how symmetry constraints, analytic field theory, and numerical optimisation combine to give a clear and quantitatively verified picture of quadrupole field quality, and they provide a self-contained computational framework that can serve as a starting point for more realistic magnet design studies.

References

- [1] H. Wiedemann, *Particle Accelerator Physics*, 4th ed. Springer, 2015.
- [2] S. Y. Lee, *Accelerator Physics*, 2nd ed. World Scientific, 2004.
- [3] E. D. Courant, M. S. Livingston, and H. S. Snyder, “The strong-focusing synchrotron – a new high energy accelerator,” *Phys. Rev.*, vol. 88, no. 5, pp. 1190–1196, 1952.
- [4] D. J. Griffiths, *Introduction to Electrodynamics*, 4th ed. Cambridge University Press, 2017.
- [5] K. H. Mess, P. Schmüser, and S. Wolff, *Superconducting Accelerator Magnets*. World Scientific, 1996.
- [6] J. A. Nelder and R. Mead, “A simplex method for function minimization,” *Comput. J.*, vol. 7, no. 4, pp. 308–313, 1965.
- [7] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. Wiley, 1999.